DATA PULSE

FMCG Analytics Project Report

End-to-End Sales & Inventory Intelligence · 2022 – 2025

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Python · EDA

SQL · ETL Pipeline

Tableau · Dashboard

Parquet + CSV Sources

Agenda

What we'll cover in this presentation

01

Business Context & Goals

Why this project was built and what it needed to deliver

02

Data Sources & Toolkit

Input files, ETL pipeline, Python, SQL, and Tableau

03

System Architecture

Stage → Transform → Load pipeline with table-level detail

04

Business Problems Identified

Key pain points the business could not answer without data

05

Analytical Approach

How each layer of the pipeline solved a specific problem

06

Key Findings & Insights

Revenue, product portfolio, and inventory discoveries

07

Recommendations

Six prioritised actions the business can take today

Business Context & Goals

Understanding why this project was built and what it needed to deliver

01

Business Context & Goals

A mid-sized FMCG company operating in 3 Polish regions with 30 SKUs across 5 categories needed to replace gut-feel decisions with data-driven intelligence.

Company Snapshot

30 SKUs	across 5 product categories
$20.3M	total revenue · 2022–2025
3 Regions	PL-Central, PL-North, PL-South
3 Pack Types	Carton · Multipack · Single
4 Lifecycle Stages	Introduction, Growth, Mature, Decline

Business Goals

Understand revenue trends

Track YoY performance and identify seasonal peaks

Optimise inventory

Reduce overstock and eliminate costly stockouts

Evaluate product portfolio

Identify top & declining SKUs by lifecycle stage

Regional analysis

Pinpoint which regions drive growth vs. lag behind

Data Sources & Toolkit

Where the data came from and the tools used to process it

02

Data Sources & Toolkit

Two categories of input files feed the ETL pipeline, processed by Python · SQL · Tableau

Input Data Files

Source : Kaggle

BATCH PARQUET FILES · via Data.feed_stage_pumpkin.py

- batch_MI-006_2020-09-06.parquet
- batch_MI-006_2020-11-13.parquet
- batch_MI-006_2020-11-20.parquet
- batch_MI-006_2020-11-27.parquet

FLAT FILES · Weekly predictions & historical sales

- dl_weekly_MI-006_prediction.csv
- FMCG_2020_2026.csv

SQL

ETL & Ingestion

- Stage → Transform → Load pipeline
- retailer & consolidated sales views
- revenue & inventory turnover CTEs
- Regional segmentation pivot queries

Python

EDA & multivariate analysis

- 3 EDA notebooks (EDA1, EDA2, EDA3)
- Data.feed_stage_pumpkin.py ingestor
- pandas · numpy · matplotlib · seaborn
- Data cleaning & feature engineering

Tableau

Interactive Dashboards

- Sales Performance dashboard
- Product Analysis dashboard
- Stock & Inventory dashboard
- Live-connected to Load layer output

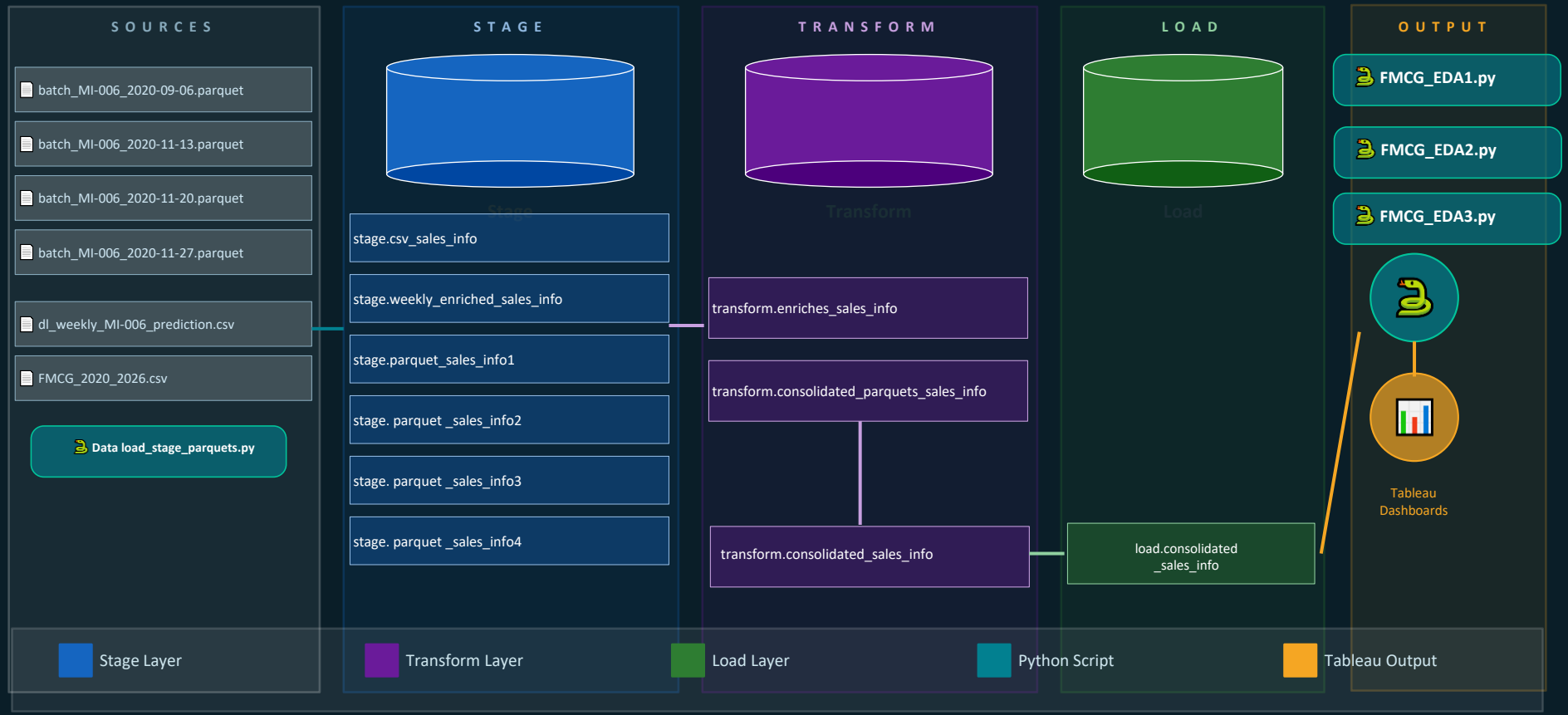
System Architecture

Stage · Transform · Load — the full ETL pipeline with table-level detail

03

ETL Pipeline Architecture

Parquet & CSV files ingest → Stage DB SQL → Transform DB SQL→ Load DB SQL → Python output → Tableau



Business Problems Faced

The core questions and pain points the business could not answer without data

04

Business Problems Identified

Four interconnected problems that required an end-to-end analytical solution

P1

No Visibility into Revenue Trends

Sales Performance

The business had no consolidated view of month-on-month or year-on-year performance. Decision-makers couldn't distinguish seasonal dips from structural decline, making budgeting unreliable.

P2

Inventory Imbalances & Stockouts

Stock & Inventory

Stock levels varied wildly across regions. Juice had the lowest inventory turnover while Yogurt peaked at 0.88 — some categories were chronically overstocked, others ran dry.

P3

Unclear Product Portfolio Health

Product Analysis

With 30 SKUs across 5 categories, 95% of units sat in the 'Introduction' lifecycle stage — raising serious questions about classification accuracy and product strategy.

P4

Regional Performance Gaps Unknown

Regional Intelligence

Stock in PL-South surged to 18.8M units (2024) while PL-Central collapsed to just 48K — severe imbalances invisible without a consolidated Stage→Load pipeline and dashboard.

Key Findings & Insights

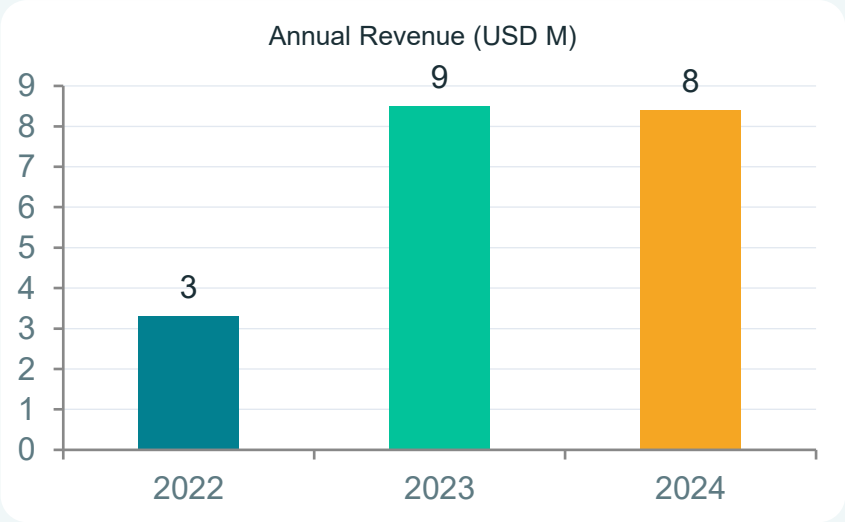
What the data revealed about sales, products, and inventory

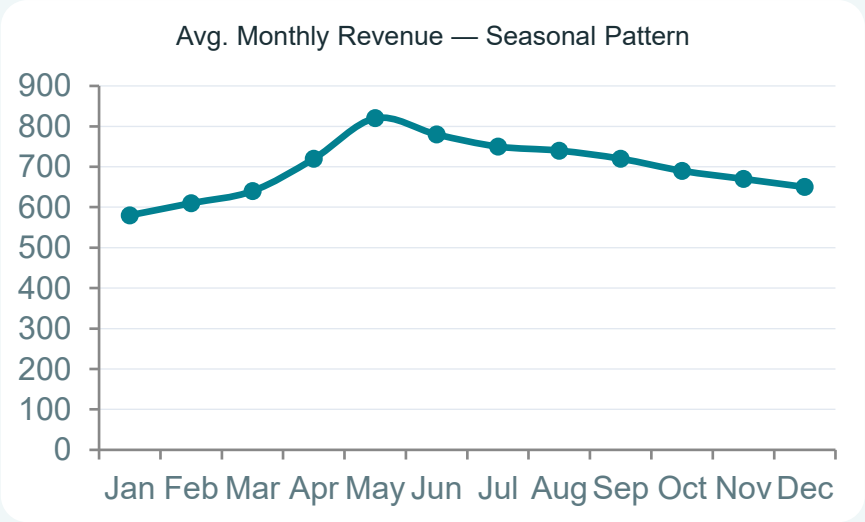
06

Finding 1 · Sales Performance

Revenue grew 157% from 2022 to 2023, then plateaued — with a consistent seasonal peak every May

$20.3M	$8.5M	+157%	+1.03%	May
Total Revenue 2022–2025	Peak Year (2023)	2022→2023 Growth	2023→2024 Growth	Peak Month Every Year

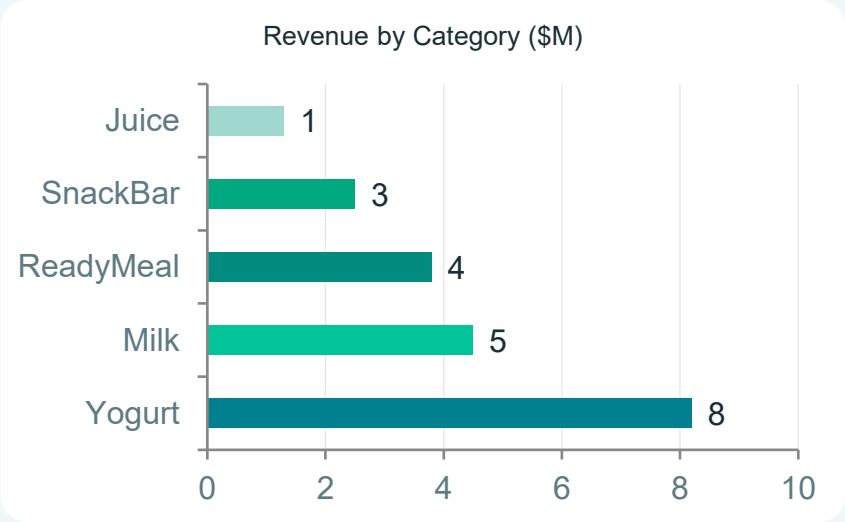


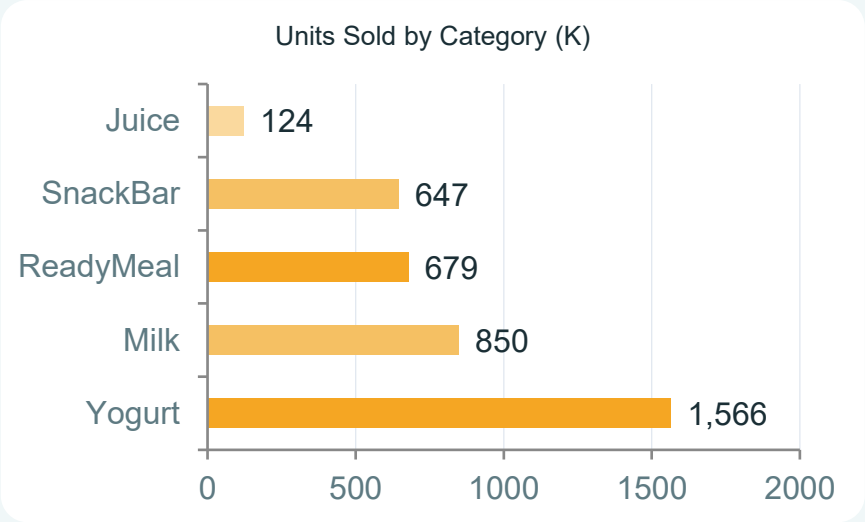


Finding 2 · Product Portfolio

Yogurt dominates both revenue and units; 95% of SKUs sit in 'Introduction' stage — a data quality concern

30	$8.2M	$2.9M	95%	~Equal
Total SKUs 5 Categories	Yogurt Revenue #1 Category	SnBrand2 Top Brand	SKUs in Intro Stage	Carton/Multi /Single Split

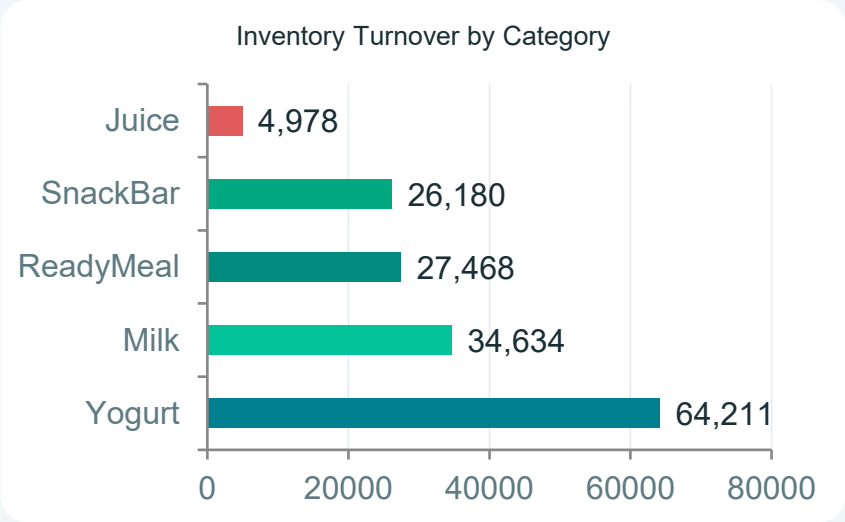


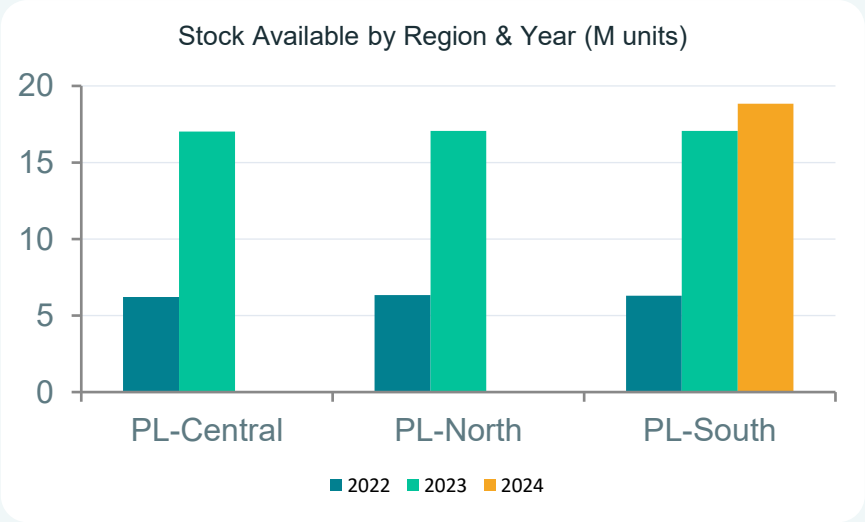


Finding 3 · Inventory & Stock

Yogurt leads inventory turnover; Juice lags — PL-South holds nearly 3× more stock than PL-Central in 2024

0.82	0.88	0.72	18.8M	48K
Avg Turnover Ratio	Yogurt (Best) Turnover	Juice (Worst) Turnover	PL-South Stock 2024	PL-Central Stock 2024 (!)





Recommendations & Next Steps

07

Turning findings into decisions the business can act on immediately

Recommendations

Six prioritised actions across sales, inventory, product strategy, and data engineering

R1 HIGH

Investigate PL-Central Stock Collapse

Central region stock fell from 17M (2023) to 48K (2024). Urgently audit the ETL pipeline at the Stage and Transform layers for data loss, then restock to avoid regional stockouts.

R2 HIGH

Capitalise on May Revenue Peak

May is consistently peak month every year. Pre-load Stage tables and coordinate supply chain 6 weeks ahead; plan promotions to maximise sell-through during the demand spike.

R3 MED

Rebalance Juice Inventory

Juice has the lowest inventory turnover (4,978 vs Yogurt 64,211). Reduce order quantities, investigate load.consolidated_sales_info for mis-allocation, and clear overstock.

R4 MED

Fix Lifecycle Stage Classification

95% of SKUs classed as 'Introduction' is implausible. Review the Transform layer logic that derives lifecycle stage and align it to actual sales velocity from stage.pamper_sales_info.

R5 LOW

Scale Yogurt & Milk Investment

Yogurt ($8.2M) + Milk ($4.5M) = 63% of total revenue. Expand SKU range, negotiate supplier capacity, and ensure the Stage→Load pipeline handles volume growth without latency.

R6 LOW

Automate the ETL & Tableau Refresh

Schedule Data.feed_stage_pumpkin.py to run daily, add data quality checks at the Transform layer, and configure Tableau Live Connection to the Load DB for real-time dashboards.

Project Deliverables

Everything produced as part of this end-to-end FMCG analytics engagement

Python Scripts & EDA

- Data ingestion script
- FMCG_EDA1/2/3.ipynb — 3 analysis notebooks

SQL ETL Pipeline

- Stage layer — 6 landing tables
- Transform layer — 3 business logic tables
- Load layer — load.consolidated_sales_info

Tableau Dashboards

- Sales Performance — trends & YoY growth
- Product Analysis — brands, lifecycle, packs
- Stock & Inventory — turnover & regional stock

This Slide Deck

- Executive-ready architecture overview
- Business problems → ETL solution → insights
- Six prioritised recommendations with rationale

DATA PULSE

Thank You

FMCG Analytics · Stage → Transform → Load · 2022–2025

$20.3M	6 Tables	3 Tables	Tableau
Revenue Analysed	Stage Layer	Transform Layer	3 Dashboards

Pipeline: Data.feed_stage_pumpkin.py · SQL ETL (Stage/Transform/Load) · Python EDA · Tableau · Data source: FMCG_2020_2026.csv + MI-006 Parquet